THE COST OF INTELLIGENCE

How large language models move from raw data to deployment—and what training consumes.

TRAINING AN LLM IS A SYSTEM, NOT A SINGLE GPU RUN

Data quality, hardware efficiency, evaluation and safety all affect cost and capability.

01
SOURCE
Web, books, code, licensed datasets

→

02
PREPARE
Filter, deduplicate, tokenize & audit

→

03
PRETRAIN
Predict tokens; update weights

→

04
ALIGN
Instruction tuning, preference feedback

→

05
EVALUATE
Test quality, bias, security & safety

→

06
DEPLOY
Serve users; monitor & iterate

FEEDBACK LOOP

Evaluation findings trigger data revisions, fine-tuning, safeguards, and retraining.

Costs also continue AFTER training: inference, hosting, monitoring, and updates.

FOUR COST DRIVERS

A consistent classification: input assets, training infrastructure, utility demand and elapsed capacity.

DATA

Collection, licensing, curation

Sources: documents, code, multilingual text;
cleaning, filtering, labeling and rights checks.

Value: quality and diversity improve coverage.

Cost: acquisition, processing and people.

COMPUTE

Accelerators, memory, networking

GPU/TPU-hours = devices × hours used;
clusters need fast interconnect and storage.

Value: enables scale and experimentation.

Cost: hardware, rentals and engineering.

ENERGY

Electricity, cooling, facilities

Power draw × operating hours estimates energy;
PUE accounts for facility overhead.

Value: keeps training infrastructure running.

Cost: kWh, cooling and carbon impact.

TIME

Wall-clock duration, iteration

Training time depends on GPUs, efficiency,
data volume, interruptions and re-runs.

Value: faster cycles speed learning.

Cost: schedule risk and opportunity cost.

REAL MODELS. DIFFERENT DISCLOSURES.

Published values are not interchangeable estimates of total company spending.

META / LLAMA 3.1 405B

~15T
pretraining tokens

30.84M
H100 GPU-hours

8,930
metric tons CO2e, location-based

Meta reports >16,000 H100 GPUs for the flagship training and an estimated emissions footprint.

Sources: Meta Llama 3.1 model card + launch post. GPU-hours ≠ elapsed hours.

DEEPSEEK / V3

14.8T
pretraining tokens

2.788M
H800 GPU-hours

$5.576M
assumed compute rental

Cost uses $2 per H800 GPU-hour; excludes earlier research, data, staffing, infrastructure ownership and ongoing serving.

OPENAI / GPT-4

Training resource totals

NOT DISCLOSED

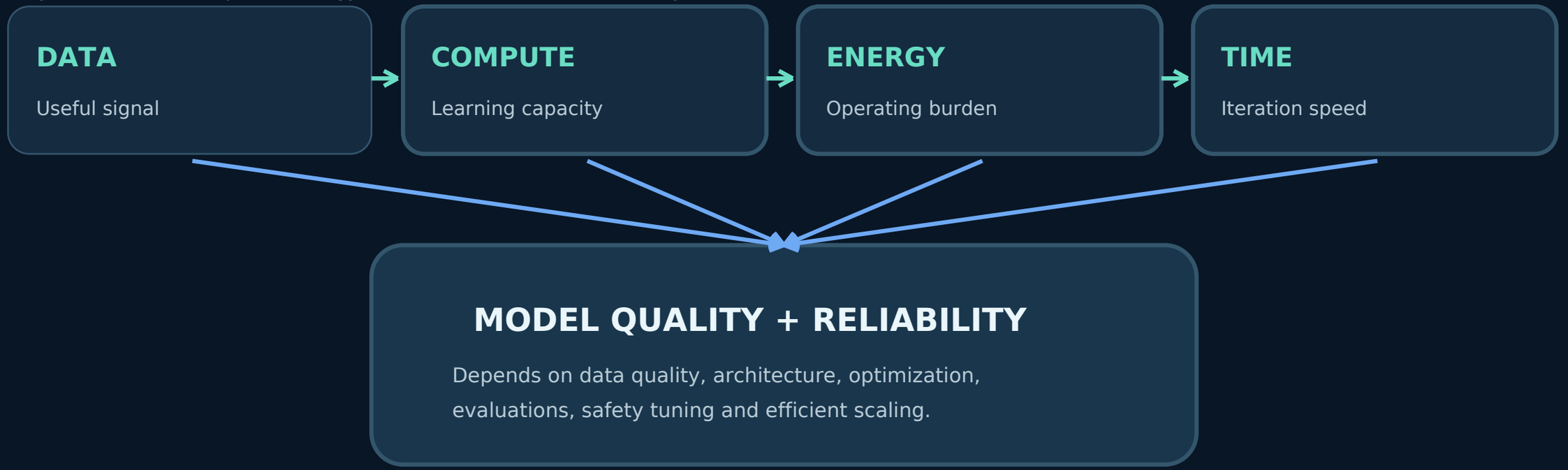
OpenAI describes next-token training and RLHF, but not comparable hardware or compute figures.

INTERPRETATION

Comparing an open model’s GPU-hours to a proprietary model’s undisclosed budget would imply false precision. Architecture, chip type, accounting scope and reporting period must align before cost comparisons.

THE RESOURCE RELATIONSHIP

Why more data, compute, energy and time do not automatically mean a better model.



COST CONTROL LEVERS

Deduplication • efficient attention • mixed precision
checkpointing • better utilization • smaller models

WHAT TO ASK BEFORE BUYING

What costs are included? Which GPUs? Which model?
How much training vs. research vs. inference?

KEY TAKEAWAYS & SOURCE NOTES

Visualizing the resource chain clarifies investment tradeoffs and prevents misleading comparisons.

01 **The workflow is iterative.**
Data preparation, pretraining, alignment, evaluation and deployment feed back into one another.

02 **GPU-hours are a measure of work.**
They combine device count and duration; they are not the same as wall-clock training time.

03 **Electricity and emissions differ.**
kWh, peak device wattage, facility overhead and location-/market-based CO2e are distinct.

04 **A reported compute bill is not total cost.**
Staff, data, earlier experiments, capex, cooling and serving may be excluded.

05 **Disclosure is uneven.**
Use disclosed figures with scope notes; label missing GPT-4 details rather than inventing them.

PRIMARY SOURCES / OPEN ACCESS

Meta (2024): ai.meta.com/blog/meta-llama-3-1/ | huggingface.co/meta-llama/Llama-3.1-70B-Instruct

DeepSeek-AI (2024): arxiv.org/abs/2412.19437 | OpenAI (2023): cdn.openai.com/papers/gpt-4.pdf